

[TITLE — 你的海报标题，8-14 个词]

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LLMs as models of human sentence processing · SoSe 2026
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Motivation

【你的文字：动机】

约 60–80 词。为什么 **overcorrection** 值得研究；BabyLM 最初的吸引力；引 Kopylova et al. (2026)。

Research question

【你的文字：研究问题】

一句话，直接从 2-page summary 搬过来。可加一句说明为什么从 BabyLM 转向了同族参数阶梯。

Hypothesis

【你的文字：假设】

约 25 词。假设 + **overcorrection** 的操作化定义 (normalized edit distance + FP 数) 。

Method

Corpus	W&I+LOCNESS v2.1 (BEA-2019), CEFR-A dev split
Sampling	1,037 → 705 after filtering → 200 items, seed 42
Filters	5–40 tokens; ≥1 non-noop edit; no punctuation- or orthography-only items; 14 degenerate items dropped
Conditions	Qwen2.5-0.5B / 1.5B / 3B-Instruct — same family, same tokenizer, only parameter count varies ; human annotator as a fourth reference point
Prompt	zero-shot: “Correct only grammatical errors in this sentence and then only output the corrected sentence.”
Decoding	greedy; 200 × 3 = 600 generations, 0 failures
Post-processing	model commentary removed by manual annotation (prefaces 7 / 15 / 0; trailing comments 0 / 2 / 5), archived in the repository
Measures	normalized word-level edit magnitude vs the source; ERRANT span-based P / R / F _{0.5} vs the gold correction
Statistics	Friedman (tie-corrected); post-hoc Wilcoxon signed-rank, Holm-corrected; rank-biserial <i>r</i> ; 4,000-resample percentile bootstrap CIs

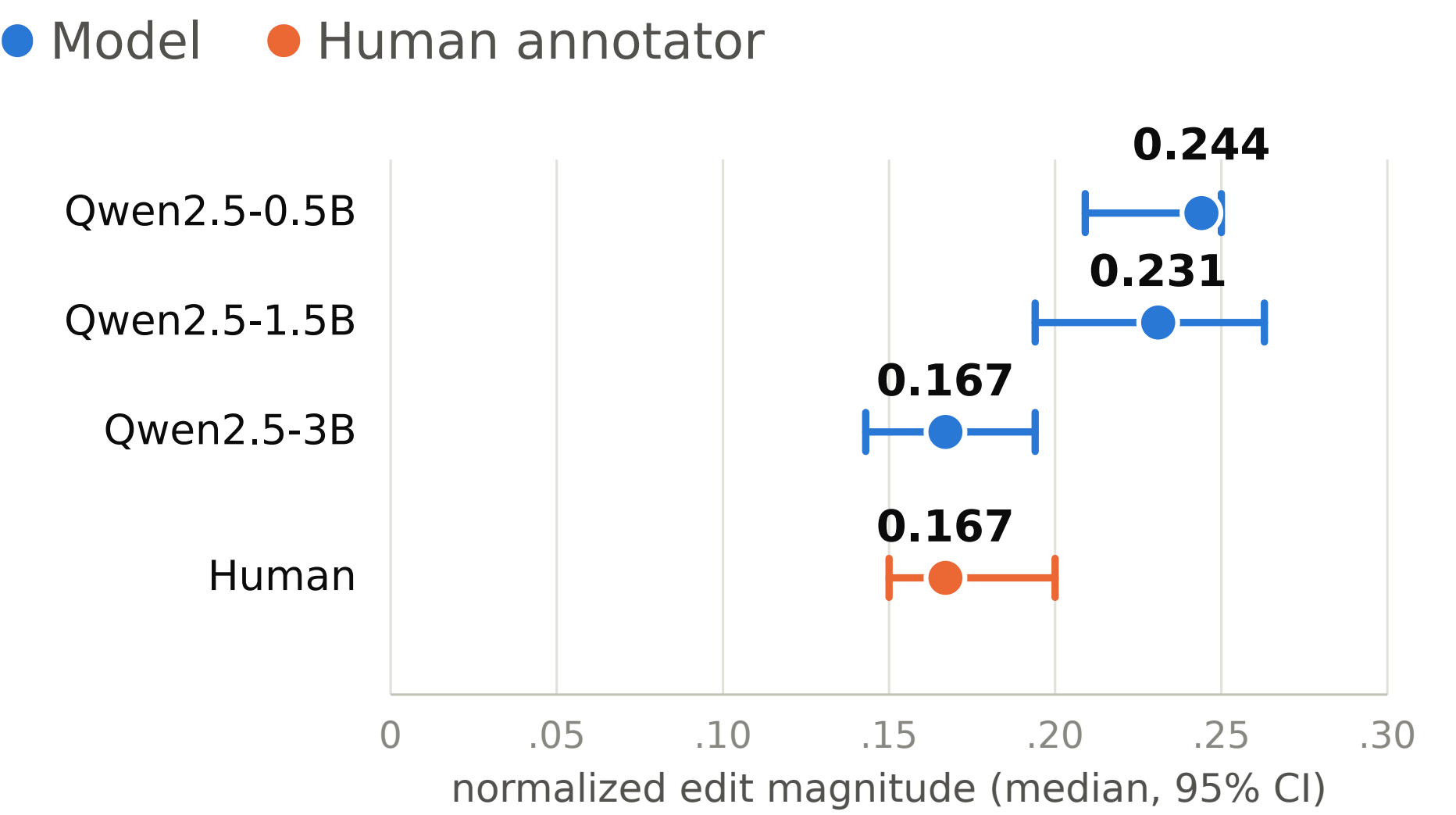
Why two measures

Edit magnitude asks **how much** a condition changed the sentence; ERRANT asks **whether the changes were the right ones**. Magnitude alone cannot separate a model that edits little because it is restrained from one that edits little because it cannot correct — a model that simply copies the input scores a perfect 0.

Finding 1 · a step, not a gradient

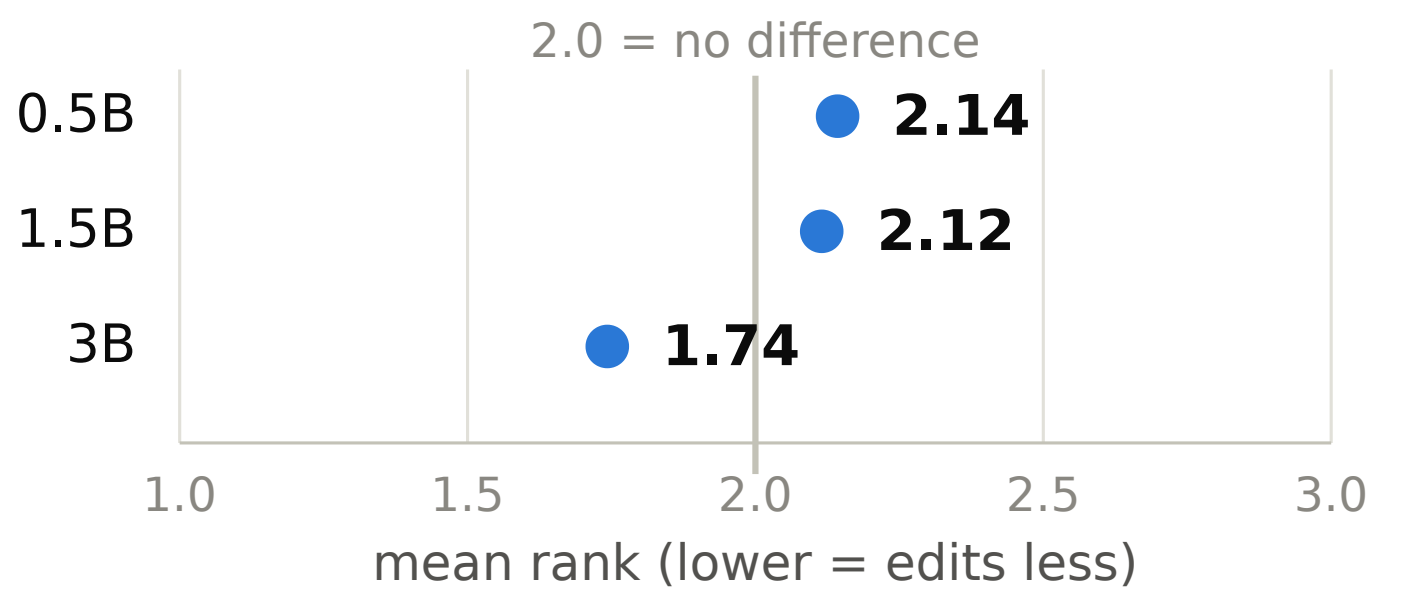
【你的文字：Finding 1 的结论】

约 30–40 词。可直接用 summary 里那句，加上“**阈值在 1.5B–3B 之间**”。



3B and the human annotator have the same median and are statistically indistinguishable ($p_{\text{holm}} = .57$, $r = .09$), while 0.5B and 1.5B do not differ from each other ($p_{\text{holm}} = .57$, $r = .10$) and both edit far more than the annotator ($r = .50$ / $.46$).

Mean rank within each sentence



Friedman $\chi^2(2) = 25.64$, $p < .001$. **0.5B and 1.5B are nearly tied; 3B sits clearly below both** — the shape of a step rather than a gradient.

One sentence, four conditions

source	It 's difficult answer at the question " what are you going to do in the future ? " if the only one who has to know it is in two minds .	edits
human	It 's difficult to answer the question " what are you going to do in the future ? " if the only one who has to know it is in two minds .	2 · .061
3B	It's difficult to answer the question "What are you going to do in the future?" if the only one who has to know it is in two minds.	2 · .061
1.5B	It's difficult to answer the question "What are you going to do in the future?" if the person who knows must be in two minds.	4 · .273
0.5B	It's challenging to answer the question "What are you going to do in the future? " when the only one who knows it is unsure .	5 · .303

Human makes exactly two minimal edits (insert to, delete at). **3B makes the same two** — its output is token-for-token the annotator's. 1.5B rewrites the subordinate clause. 0.5B also replaces difficult, if and in two minds — none of which contained a grammatical error.

Finding 2 · more fixed, less broken

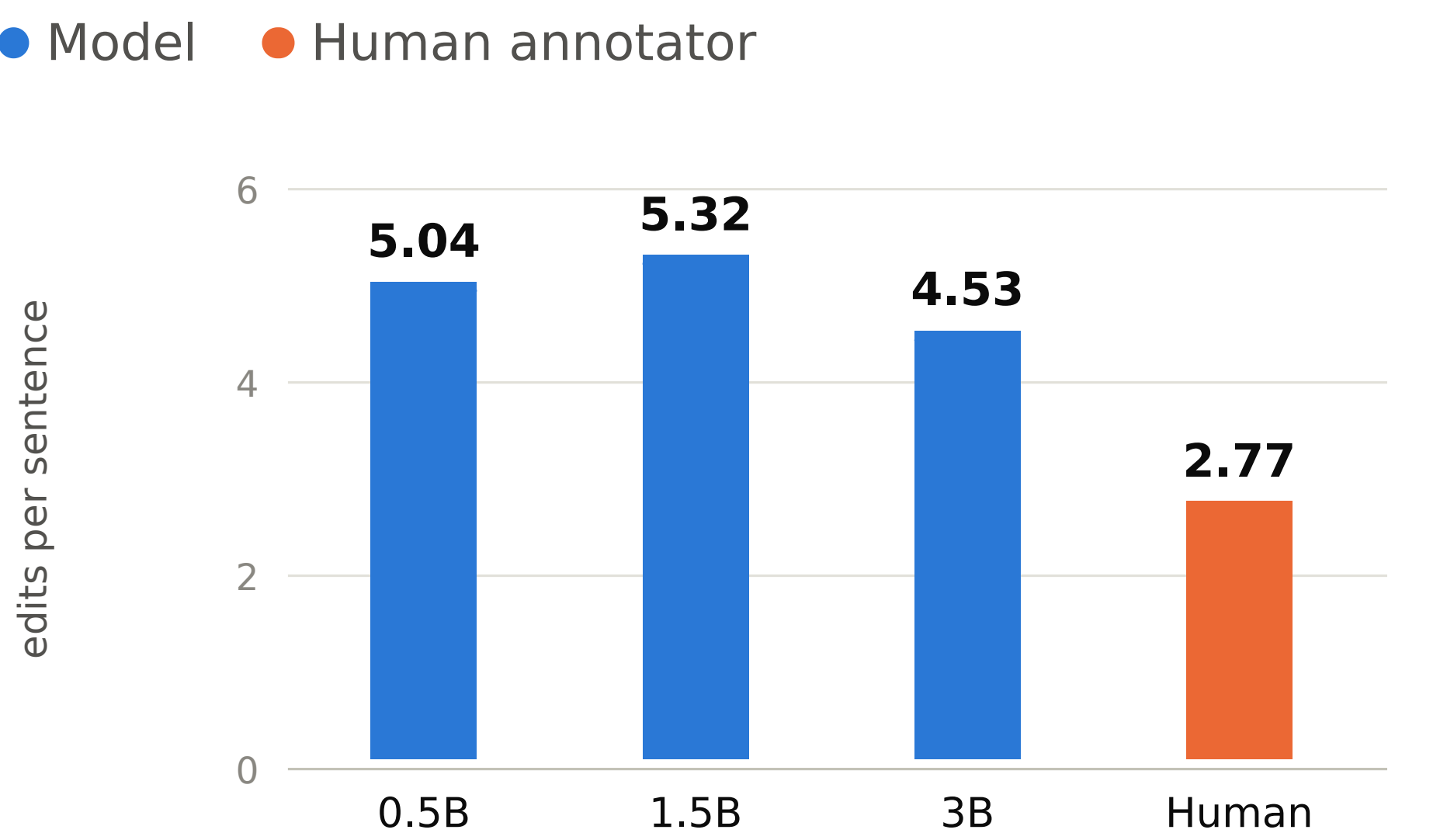
ERRANT span-based scores against the gold correction (200 items, 553 gold edits)

Model	TP	FP	FN	P	R	F _{0.5}	F _{0.5} no ORTH
0.5B	121	887	432	.120	.219	.132	.161
1.5B	163	900	390	.153	.295	.170	.208
3B	181	724	372	.200	.327	.217	.289

【你的文字：Finding 2 的解读】

约 30 词。3B 比 0.5B 多做对 60 处、少做无用 163 处；但 **precision** 仍只有 .20。

How much gets changed per sentence



Every model rewrites far more of each sentence than the annotator does. **Even 3B changes 1.6× as many spans as the human**, and its precision stays at .20 — most of what it changes, a human would have left alone.

Finding 3 · ORTH is an artefact

The ORTH category — casing and whitespace only

Model	TP	FP	FN
0.5B	7	251	12
1.5B	7	251	12
3B	7	278	12

TP and FN are **identical** across all three conditions and FP varies by 11%, while total FP varies by 24%. This category tracks output formatting, not correction behaviour. Excluding it widens 3B's false-positive advantage over 0.5B from **18%** to **30%**.

Conclusion

【你的文字：结论】

约 40–50 词。假设被推翻 (参数多的模型 **overcorrection** 更少)；台阶位置；对“本地部署 + 教学用途”意味着什么。

3B ≈ human

edit magnitude: no detectable difference ($r = .09$)

0.5B ≈ 1.5B

no detectable difference either ($r = .10$)

References

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Kopylova, E. V., Tsegoeva, O. G., Berlin, V. A., Vyrenkova, A. S., & Kuvshinskaya, Y. M. (2026). Correcting or rewriting? An expert evaluation of LLM-based GEC on academic learner data. *Dialogue 2026*, 24, 257–266.

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Code & materials

[REPO URL]

Corpus text is not redistributed (W&I+LOCNESS licence §6). The repository ships the sampling manifest (ids + seed + checksum), the manual annotation of model commentary, all analysis code, and the per-item measurement table.